

AI tooling reflection

SMG Stage 2 — accompanying note

Alongside: *Booking lifecycle and contention*

The brief asked how I directed, evaluated and corrected AI, not whether I could work from a blank page. This is that note.

I used VS Code for the source Markdown, ChatGPT for first-pass structuring and race lists, and Grok to review drafts against the brief and to pressure-test the allocation path. Gemini was used briefly for an early data-model sketch, then discarded when it started to look like a generic booking schema.

Where it was genuinely useful, and how I prompted it

AI was useful as a **structure and adversary**, once I already had an angle.

I did not prompt “design a retail media platform.” I prompted with constraints:

- the four questions from the brief;
- “availability is not a stored field”;
- “do not invent named physical slots”;
- “walk two traders taking the last unit, then expiry versus confirm, then retry after commit”;
- “keep authorised extra capacity distinct from accidental over-allocation.”

That produced:

- a stable skeleton for the four required answers;
- a first list of races I could then accept or reject;
- consistent domain language (position vs campaign, hold vs confirm, case vs incident);
- a review pass on later drafts that caught **PARTIAL** on a Hold response — which would have meant silent quantity shrink.

The useful pattern was: **state the invariant, ask what breaks it, keep the output as a draft.**

Where it was wrong, and I had to push back

Several answers were fluent and incorrect. These are the ones I would still reject in a live exercise.

Confirm as lock-free.

The model argued that because **held - 1** and **confirmed + 1** leave total consumed unchanged, Confirm need not lock **inventory_position**. That loses the aggregate pair a concurrent Hold is reading. Confirm is shorter than Hold. It is not lock-free.

One “oversell” number.

Physical overcapacity, authorised extra, uncovered overallocation and confirmed shortfall were collapsed into a single risk field. That is how a defect gets filed as a commercial decision.

Confirmed flowing back to Released.

An early state diagram allowed confirmed inventory to move back into released. The brief says a confirmed booking stands.

Silent shrink.

Given requested 4 and bookable 2, drafts offered **Hold 2** or a bulk result of **PARTIAL** as if the allocator could choose the split. The advertiser's consent is the split. The command either takes the consented quantity or writes nothing and returns a new offer.

Allowance as more physical panels.

Extra capacity was treated as changing physical capacity from 6 to 7. Physical stays 6. The allowance is separate, prior, and auditable.

Coverage as a "cryptographic" link.

A foreign key plus an audit row is enough. Calling it cryptographic is decoration.

False edit instructions.

A review pass told me to delete a Quantity-section lead-in that did not exist. I had to check the file rather than trust the comment. That is the same habit the interview will test.

I also did not let the tool decide that "exception" in the brief meant hold-stealing. I listed three readings (displace a hold / sell above physical / bend hold policy) and chose the oversell reading because the brief ties exceptions to accepting fulfilment risk. AI can enumerate. It should not close an ambiguity the brief left open.

What I deliberately did not use AI for

Choosing the angle.

Booking lifecycle and contention was my call. The tool will happily write all three example angles at equal length. Depth is a human rationing decision.

What counts as inventory truth.

Demand is not stock. Availability is a snapshot. The write transaction is authority. I would defend those without a prompt.

The handwritten working notes.

Actors, "but I really need one → exception?", and the first domain sketch were done on paper before I asked a model to tidy anything.

What to cut.

The Confirm section grew into a second paper. I cut the repeated "what Confirm is allowed to do" and the ten-bullet "what this prevents" list so Hold and Confirm were the same kind of artefact. A model will add sections. It will not spend your two hours.

Live SMG org charts.

I did not ask a model to infer the trader's employer from current careers pages. The brief left that unspecified. The design only needs: a human operator assembling demand against one retailer's estate.

How I would work in the interview

I would keep the same loop:

1. State the invariant in one sentence.
2. Ask the tool for a race or a transaction sketch against that sentence.
3. Check whether the output invents a second source of truth, a silent shrink, or a lock it cannot justify.
4. Keep or throw away by that test.

The question I care about is not whether the assistant can write Markdown. It is whether I still know what the business is allowed to promise when the last aisle barrier is gone.